MAYUR GANGIREDDY

AI/ML Engineer • Full-Stack Systems Developer

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PROFESSIONAL SUMMARY

AI/ML & Full-Stack Engineer with proven production experience deploying 5-second interval time-series pipelines, edge intelligence, and cross-platform mobile systems published to the Google Play Store and Apple App Store. Specialized in energy disaggregation (NILM), privacy-preserving Model Context Protocol (MCP) data proxies, native hardware integrations (USB IRDA / OCR), and high-throughput web microservices.

WORK EXPERIENCE

Bharat Smart Services – **Machine Learning & Full-Stack Engineer** 01/2026 – Present | Hyderabad, India

- **NILM Time-Series Energy Disaggregation:** Built production appliance disaggregation models on 5-second interval smart meter telemetry. Benchmarked 1D/2D CNNs and Temporal Transformers against an optimized sliding-window **XGBoost** pipeline; achieved **96% F1-score on Geysers** and **~85% F1-score across AC, Refrigerator, and Washing Machines** (evaluated on held-out temporal splits), yielding sub-10ms CPU inference latency.
- **NerveNet – Client-Side Encrypted Enterprise MCP Platform:** Architected an enterprise AI agent platform on the **Model Context Protocol (MCP)**. Engineered client-side AES-256 encryption and key tokenization over private CSV tables and SQL databases, allowing cloud LLMs to execute reasoning and queries strictly over anonymized tokens without exposing unencrypted corporate records.
- **Bharat Smart Services App (Utility & Billing)** Google Play App Store: Engineered cross-platform React/Capacitor app for live state discom billing (TSSPDCL, TSNPDCL, HMWSSB). Built custom native Java USB serial drivers for optical IRDA dongles, camera-based OCR meter reading review, and native iOS/Android home screen widgets.
- **Bharat Smart Home (IoT Automation & Telemetry)** Google Play App Store: Architected smart home platform featuring hierarchical device control, condition-based automation schedules, lifestyle scene orchestration, and real-time disaggregated appliance energy intelligence.

Telangana State Police Academy (TGPA) – **Android Application Developer** 03/2024 – 08/2024 | Hyderabad, India

- **Criminal Laws by TGPA** Google Play Store: Sole developer of official state police legal repository app published on Google Play. Built sub-50ms offline search over criminal penal codes using SQLite FTS, bilingual legal document viewers, and AES-encrypted local storage.

EY Global Delivery Services × AICTE – **Web Development Intern (MERN Stack)** 03/2025 – 04/2025 | Virtual

- Developed full-stack web applications using MongoDB, Express.js, React.js, and Node.js. Built scalable RESTful APIs, JWT authentication, and responsive administrative dashboards.

Acmegrade – **Android App Development Intern** 11/2023 – 01/2024 | Remote

- Developed and deployed native Android mobile application modules in Java, implementing activity lifecycles, UI layout hierarchies, REST API integration, and local SQLite caching.

SELECTED PROJECTS (CLIENT PRODUCTION)

- **Temple Donation Signage Hub & Devotee Android TV Network:** Devotional donation kiosk & digital signage TV network. Built administrative portal generating dynamic zero-transaction-fee UPI QR codes with automated verification polling, and a native Android TV app (Kotlin, Jetpack Compose, ExoPlayer/media3) featuring 6-character claiming codes and resilient offline media caching; deployed live on temple donation kiosks and display screens.
- **Hostel Hub Pro & Companion Mobile App:** Multi-tenant accommodation platform connecting administrators, wardens, students, and guards. Built administrative web portal with **TanStack Start SSR & React**, companion **Flutter app** featuring time-restricted (24-hr) HMAC dynamic QR gate passes with server-side nonce verification to prevent pass re-use and sharing, and an Express/MySQL backend on Plesk.
- **SignageHub – Hybrid Digital Signage & TV Network:** Commercial digital signage platform featuring cloud media scheduling dashboard and compiled local server broadcasting UDP discovery packets to auto-pair native Android TV screens. Engineered Android TV client in Jetpack Compose, Ktor, and local SQLite caching for continuous offline playlist playback with zero manual IP setup.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), Java, Kotlin, SQL, Dart, C, C++

AI & Machine Learning: XGBoost, PyTorch, Transformers, CNNs, Time-Series NILM, Model Context Protocol (MCP), Computer Vision (OCR), Scikit-Learn

Web & Mobile: React.js, Node.js, Express, FastAPI, Capacitor.js, Flutter, Android (Jetpack Compose/Java), Ktor, Tailwind CSS

Databases & Cloud: PostgreSQL, MySQL, MariaDB, SQLite, DuckDB, Supabase, Firebase, Plesk Linux, RESTful APIs

Tools & Hardware: Git, Docker, Android Studio, Xcode, Postman, Linux CLI, USB Serial Hardware (IRDA), UDP Discovery

EDUCATION & HONORS

Sphoorthy Engineering College – B.Tech in Computer Science & Engineering (AIML) 2021 – 2025 | Hyderabad, India

Sri Gayatri Junior College – Board of Intermediate Education (MPC) 2019 – 2021 | Hyderabad, India

Grand Finalist – Smart India Hackathon (SIH 2022)

Production Systems – 3 Published Apps on Google Play & App Store | Live Paid Client Deployments